# Tangents

**1** Find the gradient of the tangent to $f(x) = x^3 + 5x$ at the point $(2, 18)$.

**2** Consider the function $y = \frac{1}{3}x^3 + \frac{1}{6}x^6 + 2x$.

**a** Find $\frac{dy}{dx}$.

**b** Evaluate $\frac{dy}{dx}$ when $x = -7$.

**3** Find the gradient of $f(x) = \frac{11}{x} + \frac{10}{x^2}$ at the point $\left(4, \frac{27}{8}\right)$.

**4** Consider the parabola $f(x) = x^2 + 3x - 10$.

**a** Find the $x$-intercepts.

**b** Find the gradient of the tangent at the positive $x$-intercept.

**5** Consider the function $y = 4x^2 - 5x + 2$.

**a** Find $\frac{dy}{dx}$.

**b** Find the value of $x$ at which the tangent to the parabola is parallel to the $x$-axis.

**6** Consider the tangent to the curve $f(x) = -x^3$ at the point $(2, -8)$.

**a** Find the gradient of the function $f(x) = -x^3$ at the point.

**b** Hence, find the equation of the tangent to the curve at the point.

**7** Consider the tangent to the curve $f(x) = 5\sqrt{x}$ at the point $\left(\frac{1}{9}, \frac{5}{3}\right)$.

**a** Find the gradient of the function $f(x) = 5\sqrt{x}$ at the point.

**b** Hence, find the equation of the tangent to the curve at the point.

**8** Consider the tangent to the curve $f(x) = 6\sqrt{x}$ at the point $(4, 12)$.

**a** Find the gradient of the function $f(x) = 6\sqrt{x}$ at the point.

**b** Hence, find the equation of the tangent to the curve at the point.

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**9** Consider the tangent to the curve $f(x) = \dfrac{2}{x^3}$ at $x = -2$.

- **a** Find the gradient of the function $f(x) = \dfrac{2}{x^3}$ at $x = -2$.
- **b** Find the $y$-coordinate of the point of intersection between the tangent line and the curve.
- **c** Hence, find the equation of the tangent to the curve at $x = -2$.

**10** Consider the tangent to the curve $f(x) = -2x^2 + 8x + 2$ at $x = 1$.

- **a** Find the $y$-coordinate of the point of intersection between the tangent line and the curve.
- **b** Find the equation of the tangent to the curve at $x = 1$.

**11** Consider the tangent to the curve $f(x) = 3x^3$ at $x = 2$.

- **a** Describe the steps required to find the equation of the tangent line at $x = 2$.
- **b** Hence, find the equation of the tangent to the curve.

**12** Consider the function $f(x) = \dfrac{9x + 4}{3x}$.

- **a** Find the $y$-coordinate of the point on the curve at $x = -1$.
- **b** Hence, find the equation of the tangent to the curve at $x = -1$.

**13** Consider the function $g(x) = \dfrac{8x^7 - 6x^6 + 4x^5 + 7}{2x^2}$.

- **a** Find the derivative of the function.
- **b** Find the equation in of the tangent to the curve at $x = 1$.

**14** Find the equation of the tangent to $f(x) = x^2 + x$ at the point $(2, 6)$.

**15** Find the equation of the tangent to $f(x) = (3x - 1)(2x - 5)$ at the point $(2, -5)$.

**16** Consider the function $f(x) = x^3 - 6x^2$. Find the coordinates of the points on the curve where the gradient is $495$.

**17** Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$. Find the coordinates of the points on the curve where the gradient is $5$.

**18** Consider the function $f(x) = 2x^2 - 216\sqrt{x}$. Find the coordinates of the point on the curve

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**19** Consider the function $f(x) = \frac{4x^3}{3} + \frac{5x^2}{2} - 3x + 7$. Find the $x$-coordinates of the points on the curve whose tangent is parallel to the line $y = 3x + 7$.

**20** At point $M$, the equation of the tangent to the curve $y = x^2$ is given by $y = 4x - 4$. Find the coordinates of $M$.

**21** At point $M$, the equation of the tangent to the curve $y = x^3$ is given by $y = 12x - 16$. Find the coordinates of $M$.

**22** Find the equation of the tangent to the parabola $y = 2x^2 + 8x - 5$ at the point where the gradient is $0$.

**23** Consider the curve given by the function $f(x) = x^2 - 4x + 2$.

- **a** Find the gradient of the tangent to the curve at the point $(3, -1)$.
- **b** Find the coordinates of the vertex of the parabola $f(x)$.
- **c** Sketch the graph of the curve and the tangent at the point $(3, -1)$.

**24** Consider the curve given by the function $f(x) = x^2 - 1$.

- **a** Find the gradient of the tangent to the curve at the point $(1, 0)$.
- **b** Sketch the graph of the curve and the tangent at the point $(1, 0)$.

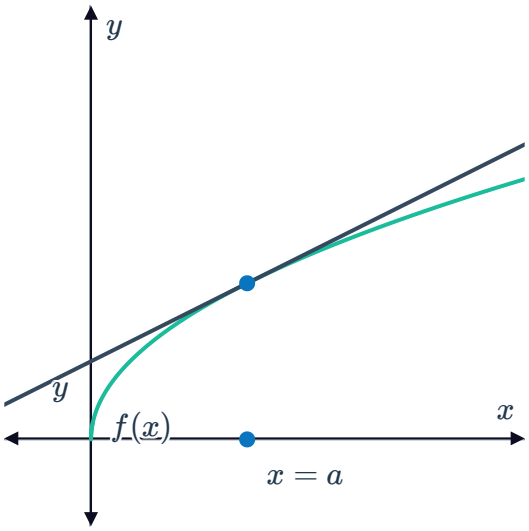
**25** Consider the curve $y = x^3 - x^2$ and the line $4x - y = 11$.

- **a** Find the $x$-coordinates of the points on the curve at which the tangents are perpendicular to the line $4x - y = 11$.
- **b** Find the equation of the tangent to the curve at each of the $x$-coordinates found in part (a).

**26** The curve $f(x) = k\sqrt{x} - 5x$ has a gradient of $0$ at $x = 16$. Find $k$.

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**27** In the figure, the straight line $y = \frac{1}{10}x + b$ is tangent to the graph of $f(x) = 6\sqrt{x}$ at $x = a$.




**a** Find $a$.

**b** Find $b$.

**28** $5x + y + 2 = 0$ is the tangent line to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.

**a** Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.

**b** State the gradient of the tangent to the curve at $x = 9$.

**c** Find the value of $b$.

**d** Find the value of $c$.

**29** The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of $58$ at the point $(2, 31)$.

**a** Use the given information to write two equations expressing $b$ in terms of $a$.

**b** Hence, find the value of $a$ and $b$.

**30** The graph of $y = ax^3 + bx^2 + cx + d$ has the following properties:

- It intersects the $y$-axis at $y = -28$, where the tangent is parallel to the $x$-axis.
- It intersects the $x$-axis at $(2, 0)$, where it has a gradient of $36$.

**a** Use the information given in the first point above to find the value of $c$ and $d$.

**b** Use the information given in the second point above to write two equations expressing $b$ in terms of $a$.

**c** Hence, find the value of $a$ and $b$.

**31** From an external point $(3, 2)$, two tangents $L_1$ and $L_2$ are drawn to the curve $y = x^2 - 6$.

**a** Let the gradient of a tangent line to the curve be $m$. Find the two possible values of $m$ that correspond to the gradients of $L_1$ and $L_2$.

**b** If $L_1$ is the tangent with the larger gradient, find the equation of $L_1$.

**c** Find the equation of $L_2$.

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## Normals

**32** Consider the function $y = 10x^5 - 2x^4 + 8x^3 - 6x^2 + 4x - 164$.

- **a** Find $\frac{dy}{dx}$.
- **b** Find the gradient of the tangent to the curve at the point $(3, 2278)$.
- **c** Find the gradient of the normal to the curve at the point $(3, 2278)$.

**33** The gradient of the curve $f(x)$ at $x = 3$ is $2$.

- **a** Find the gradient of the tangent line at $x = 3$.
- **b** Find the gradient of the normal line at $x = 3$.

**34** Consider the curve $f(x) = x^2 + 8x + 15$ at the point $(4, 63)$.

- **a** Find $f'(x)$.
- **b** Find the gradient of the tangent to the curve at the point.
- **c** Find the equation of the tangent to the curve at the point.
- **d** Find the gradient of the normal to the curve at the point.
- **e** Find the equation of the normal to the curve at the point.

**35** Consider the curve $f(x) = 4x + \frac{64}{x}$ at the point $(4, 32)$.

- **a** Find the equation of the tangent to the curve at the point.
- **b** Find the equation of the normal to the curve at the point.

**36** $-4x + y + 1 = 0$ is the normal to the curve $y = x^2 + bx + c$ at the point $(-8, -32)$.

- **a** Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.
- **b** State the gradient of the normal to the curve at $x = -8$.
- **c** Find the value of $b$.
- **d** Find the value of $c$.

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## Applications

**37** Consider the curve $f(x) = 36 - x^2$ at the point $P(3, 27)$.

- **a** Find $f'(x)$.
- **b** Find the gradient of the tangent to the curve at the point.
- **c** Find the equation of the tangent to the curve at the point.
- **d** Find the gradient of the normal to the curve at the point.
- **e** Find the equation of the normal to the curve at the point.
- **f** The tangent line at $(3, 27)$ meets the $x$-axis at point $A$. Find the $x$-coordinate of $A$.
- **g** The normal line at $(3, 27)$ meets the $x$-axis at point $B$. Find the $x$-coordinate of $B$.
- **h** Find the length of $AB$.
- **i** Find the exact area of $\triangle ABP$.

**38** The normal to the curve $y = x(x-5)^2$ at the point $A(6, 6)$ cuts the $x$-axis at $B$.

- **a** Find the gradient function.
- **b** Find the gradient of the tangent at $A$.
- **c** Find the gradient of the normal at $A$.
- **d** Hence, find the $x$-coordinate of point $B$.
- **e** Find the area of triangle whose vertices are the origin, point $A$ and point $B$.

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